	GLOBAL ECONOMICS **GRADE: 11TH** **TERM 3 — LEARNING EVIDENCE 1** **HYPOTHESIS TEST DESIGN (C1–C3)** **TEACHER'S NAME: Nicolás López Cuéllar**	**THIRD TERM 2025–2026**

Learning objective: Design hypothesis tests from real contexts by identifying whether the parameter is a population mean or proportion, defining and sampling in context, setting and justifying a benchmark claim, stating a significance level, and determining the test tail direction from the contextual claim.	**Evaluated criteria:** C1: Describes the concept of a statistical hypothesis. C2: Identifies the null and alternative hypotheses in a statistical test. C3: Determines whether a test should be one-tailed or two-tailed.

Criteria assessment

Each assessment criterion is evaluated through the worked examples and activity questions. A criterion is considered passed when it is correctly applied in 4 of the 6 assigned tasks.

For a set of tasks to count as valid and complete, it must include

- exactly 3 proportion/probability settings and exactly 3 mean settings, so that Criterion C1 can be assessed properly
- at least 1 left-tailed setup (lower than), at least 1 right-tailed setup (higher than), and at least 1 two-tailed setup (different from), so that Criteria C2 and C3 can be assessed properly

Flashcards with the meaning of the different symbols are allowed. However, all flashcards will be checked. If a flashcard contains a solved or developed problem, the exam will be nullified. In addition, no exams will be accepted after the established submission time. Any late submission, even if it is only 5 seconds late, will not be received.

Introduction

A hypothesis test starts with a contextual benchmark claim about a population. In this activity, you will design that claim yourself: first decide whether your context is about a population mean or a population proportion, then define the parameter, collect or propose sample information, justify a benchmark, state a significance level, and translate the claim into formal hypotheses. The main goal is to build the structure of the test clearly and justify the tail direction from the wording of the alternative claim.

General problem. A benchmark claim may refer either to a population proportion or to a population mean, and in this activity you will create the benchmark from the context.

For a proportion setting, define a population characteristic that is measurable as success/failure and let the parameter be the true population proportion p. Propose a plausible benchmark value p_0, explain why that benchmark makes sense in your context, and state a significance level.

For a mean setting, define a measurable quantitative variable and let the parameter be the true population mean μ. Propose a plausible benchmark value μ_0, explain why that benchmark is reasonable for your context, and state a significance level.

In both settings, describe how sample information would be collected, including a sample plan or sample size, invent a plausible sample outcome that could be compared with the benchmark, and justify the test direction from the wording of the claim.

Questions/Tasks.

(C1) For your chosen context, identify whether it is a proportion setting or a mean setting; define the parameter in context; explain how sample information would be collected; and state a plausible sample result.

(C2) State the benchmark claim in context; justify the benchmark narratively; state a significance level; then write the null and alternative hypotheses with correct notation, keeping the null as "parameter equals benchmark."

(C3) Classify the test as left-tailed, right-tailed, or two-tailed, and justify the classification from the wording and intention of the claim.

Example 1

Slippers logistics for the school event

The school is preparing for a large event. For logistics, students in Grades 8 to 11 are treated as the "grown ups" in charge of extra support: each one is asked not only to bring their own slippers, but also two extra pairs to share with students in Grades 4 to 7.

Because of this, the organizing team wants to know whether the average number of slippers brought per person by students in Grades 8 to 11 is greater than 4. A survey is planned by going classroom to classroom on one day. However, many students are assigned to other event activities that same day, so not everyone is available to answer. The expected sample is 120 students, and a plausible proposed sample outcome is $\bar{x} = 4.5$ slippers per student.

The organizers formulate this as a hypothesis test with significance level $\alpha = 0.05$.

C1

- This is a population mean setting.
- Let μ be the true population mean number of slippers brought per student among Grades 8 to 11.
- Sample information would be collected by visiting classrooms and recording each available student's reported number of slippers brought that day.
- Because some students are busy in parallel event duties, the survey is expected to reach about $n = 120$ students rather than all eligible students.
- The proposed sample outcome is $\bar{x} = 4.5$ slippers per student.

C2

- Benchmark claim in context: the average number of slippers brought is 4 per student, so $\mu_0 = 4$.
- Benchmark justification: bringing one personal pair plus two extra pairs suggests 3 as a minimum personal target, but planners set 4 as a safety benchmark to protect younger grades when participation is uneven.
- Significance level: $\alpha = 0.05$.
- Hypotheses: $H_0\colon \mu = 4$, $H_a\colon \mu > 4$.

C3 Right-tailed test, because the key claim is "greater than 4," and the direction matches the practical question of whether average slipper availability exceeds the benchmark level.

Example 2

Passing exams without studying (diary records)

A student keeps a personal diary with a very consistent method: after each exam attempt, they record whether they passed without studying. They claim they "never study," so they have accumulated plenty of personal data over time.

In this course, passing all requirements means passing 7 criteria, and each criterion allows 3 opportunities. The student wants to know whether their true probability of passing an exam attempt without studying is $\frac{1}{3}$ or lower. If the probability is at or below $\frac{1}{3}$, that would be a strong signal that studying is necessary to avoid failing the course overall. From the diary, a plausible sample is $n = 84$ recorded attempts with $x = 24$ passes (so $\hat{p} = 24/84 \approx 0.286$). The test is formulated with significance level $\alpha = 0.10$.

C1

- This is a population proportion setting.
- Let p be the true probability that this student passes an exam attempt without studying.
- Sample information comes from the student's diary records, where each attempt is coded as pass/fail.
- A plausible sample summary is $n = 84$ attempts with $x = 24$ passes, giving $\hat{p} \approx 0.286$.

C2

- Benchmark claim in context: $p_0 = \frac{1}{3}$.
- Benchmark justification: each criterion has 3 opportunities, so $\frac{1}{3}$ functions as a practical minimum success benchmark per attempt when judging whether the "never study" strategy can survive the full course structure.
- Significance level: $\alpha = 0.10$.
- Hypotheses: $H_0\colon p = \frac{1}{3}$, $H_a\colon p < \frac{1}{3}$.

C3 Left-tailed test, because the concern is whether the true passing probability is lower than the benchmark, and that direction directly matches the question "is $\frac{1}{3}$ or lower?" in decision form.

Example 3

Bogotá "ya llego" arrival-accuracy audit

A Grade 11 group in Bogotá notices a recurring urban ritual: when someone texts "ya llego" ("I'm arriving now"), classmates quietly assume that "now" is not literal. Instead of arguing in the chat every

Friday, they decide to treat the issue with formal seriousness.

They define the measurable variable as waiting time (in minutes) from the moment a person sends "ya llego" to the moment that person actually arrives at the meeting point (for example, near a TransMilenio station entrance). The group sets 8 minutes as the reference for what that phrase should mean in practice, and they want to check whether the true average is different from that benchmark.

In their logic, both directions matter: if the average is far above 8 minutes, "ya llego" sounds optimistic; if it is far below 8 minutes, people may be sending the message only when they are basically at the corner already. Over several weekends, they plan to record 60 such cases from volunteer classmates and trusted friends in Bogotá, across different zones and traffic conditions. A plausible proposed sample outcome is $\bar{x} = 9.6$ minutes. They take $\alpha = 0.05$.

C1

- This is a population mean setting.
- Let μ be the true mean waiting time (in minutes) between sending "ya llego" and actual arrival in this Bogotá student context.
- Sample information would be collected by logging timestamp pairs: message-sent time and arrival time for each observed case.
- The sample plan uses $n = 60$ recorded "ya llego" cases from weekend meetups in Bogotá.
- The proposed sample outcome is $\bar{x} = 9.6$ minutes.

C2

- Benchmark claim in context: $\mu_0 = 8$ minutes as the reference meaning of "ya llego".
- Benchmark justification: in Bogotá, short delays from traffic lights, station queues, or quick route changes are normal, so 8 minutes is used as a practical middle reference; the audit question is whether real behavior departs from that reference, not only whether it is larger.
- Significance level: $\alpha = 0.05$.
- Hypotheses: $H_0\colon \mu = 8$, $H_a\colon \mu \neq 8$.

C3 Two-tailed test, because the claim asks whether the true average wait after "ya llego" is different from 8 minutes. In context, both departures matter: averages above 8 suggest chronic extra waiting, while averages below 8 suggest the phrase is sent much later than expected. So the question is about being either higher or lower than the benchmark, not only one side.

Student Problems

Use the following shared instructions for all six problems:

(C1) **Describe the context**; state whether it is a population mean or population proportion setting; **match the parameters** with their corresponding symbols; **explain how you would collect sample information**; and invent a plausible sample result.

(C2) **State the benchmark claim** and **justify it narratively**; choose a significance level; then write H_0 and H_a with correct notation, keeping the null as "parameter equals benchmark."

(C3) **Classify your test** as left-tailed, right-tailed, or two-tailed, **draw the corresponding graph**, and **justify the classification** from the wording and intention of your claim.

Across your six problems, your set must be balanced and directionally complete: include half proportion/probability settings and half mean settings, and include at least one lower-than claim, at least one higher-than claim, and at least one different-from claim.

1. **Problem 1 — School context**
   Create a hypothesis test about a measurable feature of your school.
2. **Problem 2 — Home context**
   Create a hypothesis test about a measurable feature of life at home.
3. **Problem 3 — Neighborhood context**
   Create a hypothesis test about a measurable feature of your neighborhood.
4. **Problem 4 — Your Grade 11 group context**
   Create a hypothesis test about a measurable feature of your own Grade 11 group.
5. **Problem 5 — Personal context**
   Create a hypothesis test about a measurable feature related to yourself.
6. **Problem 6 — Bogotá context**
   Create a hypothesis test about a measurable feature of Bogotá that can be investigated at your level.